Expanding Synthesizable Chemical Space: Benchmarking Post-Generation Molecular Modification

Tyler Biswurm[1*], Jingxin Huang[1], Yuma Nakamura[1], Ben Luo[1]

[1*]Department of Computer Science, University of Toronto, Toronto, Canada.

*Corresponding author(s). E-mail(s): tyler.biswurm@mail.utoronto.ca; Contributing authors: jingxin.huang@mail.utoronto.ca; yuma.nakamura@mail.utoronto.ca; ben.luo@mail.utoronto.ca;

Abstract

Synthesis-aware generative models like PrexSyn guarantee synthesizability but are constrained by their building-block libraries and reaction templates. We investigate whether post-generation modification methods (CReM, mmpdb, JT-VAE, LibINVENT) provide measurable value in expanding chemical coverage beyond repeated PrexSyn sampling. This work establishes the first systematic benchmark of modification methods as complements to synthesis-aware generation, with complementarity analysis to inform pipeline design for drug discovery.

Keywords: molecular generation, synthetic chemistry, drug discovery, machine learning

Research Question & Significance

Synthesis-aware generative models guarantee that every proposed molecule can be assembled from purchasable reagents via validated reactions, a property critical for practical drug discovery [1]. PrexSyn [2], the current state of the art, generates synthesis pathways conditioned on desired molecular properties, producing candidates synthesizable by construction within the Enamine REAL chemical space [3]. It is fast (∼0.26 s per target) and reconstructs 94% of molecules drawn from Enamine

REAL, but only 28% of the broader ChEMBL bioactive distribution [2]. Substantial pharmacologically relevant chemistry lies beyond its reach.

This coverage boundary is inherent to the paradigm: any synthesis-aware generator is bounded by its building block library and reaction templates. Retraining on a larger space shifts the boundary; repeated sampling draws from the same constrained space, yielding more candidates but no new chemical territory.

We investigate whether classical molecular modification methods (fragment substitution, matched molecular pair transforms, latent-space editing, scaffold-constrained decoration) provide measurable value when composed with the frontier generative model. These methods predate generative AI, operate outside Enamine REAL, and are available as open-source tools. We benchmark four such methods on their ability to produce synthesizable, property-conserving candidates beyond what repeated PrexSyn sampling achieves. Methods are ranked by marginal value and stratified by how well PrexSyn performs alone, revealing when modification adds the most value. Pairwise complementarity analysis identifies which combinations maximize total coverage, informing pipeline design and tool selection for agent-driven molecular design.

Context & Datasets

PrexSyn [2] generates synthesis pathways from ∼480K Enamine in-stock building blocks via 115 reaction templates, conditioned on molecular property specifications. We use it as a black-box generator: given a property profile, it returns synthesizable candidates within Enamine REAL.

ChEMBL [4] provides realistic property specifications. We extract computed property profiles from sampled bioactive molecules and use these as input to PrexSyn.[1] The molecules themselves are not optimization targets; structural similarity to the ChEMBL source is reported only as a supplementary diagnostic.

AiZynthFinder [5], trained on the USPTO reaction corpus [6], provides retrosynthetic analysis as our synthesizability gate. Each generated variant is checked for decomposability into purchasable Enamine building blocks via known reactions.

The four modification methods draw on independent resources: CReM [7] uses a context-constrained fragment database mined from ChEMBL; mmpdb [8] applies matched molecular pair transformation rules; JT-VAE [9] encodes and decodes molecules via a learned latent space; LibINVENT [10] decorates molecular scaffolds via reinforcement learning (accessed through REINVENT4 [11]).

[1] Exact specification format (ECFP4 fingerprints, natively accepted by PrexSyn, vs. the full descriptor set from PrexSyn Table 3) is under investigation and will be finalized before experiments begin.

Proposed Methods

PrexSyn generates an initial synthesizable candidate for each property specification. Four modification methods then produce structural variants from this seed. All variants, along with a repeated-PrexSyn-sampling baseline, pass through identical synthesizability and property-conservation filters. The baseline tests whether simply drawing more samples from the generative model matches or exceeds the value of a post-generation modification layer.

CReM [7] performs mutate, grow, and link operations using a context-constrained fragment database mined from ChEMBL. A tunable context radius controls edit scale: small radii produce conservative R-group substitutions; larger radii enable ring-system and scaffold-level changes. We benchmark CReM at multiple context radii (1, 2, 3) to characterize the locality–quality tradeoff within a single method.

mmpdb [8] mines and applies single-site matched molecular pair transformations (e.g., methyl → chloro, phenyl → pyridyl). Rules carry empirical property-change annotations, enabling selection of transformations likely to preserve the desired profile. It is the standard tool for conservative, chemist-interpretable local edits.

JT-VAE [9] is a variational autoencoder over junction-tree decompositions of molecular graphs. The seed is encoded into a continuous latent space; perturbation of the latent vector and decoding produces variants spanning medium-to-global edit scales.

LibINVENT [10, 11] preserves a molecular scaffold extracted from PrexSyn's output while exploring alternative R-group decorations via reinforcement learning steerable toward target properties.

These methods span the learned/rule-based and local-to-global axes (mmpdb → CReM → JT-VAE, LibINVENT), so the benchmark is informative regardless of which strategy dominates. No prior work systematically benchmarks modification methods as post-generation complements to synthesis-aware generation. This study provides the first such comparison, including the first complementarity analysis for identifying which combinations maximize coverage.

The complementarity analysis also provides the empirical basis for composing effective operators into multi-stage pipelines, chaining local edits, fragment substitutions, and latent-space exploration with synthesis filtering at every stage. No public system yet integrates these operators under unified synthesis constraints; our rankings and complementarity matrices provide the foundation for such integration.

Data and Preprocessing

We sample 100 molecules from ChEMBL with pre-stratified difficulty: PrexSyn is first run on a larger pool to compute baseline quality scores (ECFP4 Tanimoto between best candidate and specification), then ∼25 molecules are drawn from each of four quality bins (<0.5, 0.5–0.7, 0.7–0.85, 0.85–1.0). This ensures balanced representation across difficulty levels.

For each specification, PrexSyn generates 256 candidates; the highest-scoring candidate ($K = 1$) seeds all modification methods, so results reflect the modification method rather than seed diversity. Each method produces up to N variants; N is set

by a pilot saturation study (5 specs, $N = 1,000$ per method) to identify the diversity plateau. Expected cap: 100–500. Variants are deduplicated by canonical SMILES before synthesizability checking via AiZynthFinder at search depth 1, with depth-2 sensitivity analysis on a 10% subsample. Estimated total load after deduplication: ∼80K unique molecules, ∼8 core-hours.

No models are trained in this study; all methods are used as pre-trained or pre-configured tools. CReM's fragment database derives from ChEMBL, the same source as our specifications. Since ChEMBL molecules serve only as property-profile sources, not reconstruction targets, this overlap reflects realistic tool usage rather than information leakage.

Information Flow Diagram

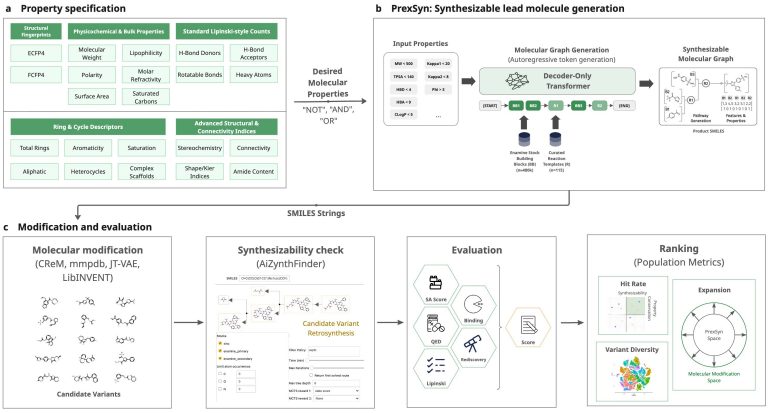

Fig. 1 (a) Users define target properties via molecular descriptors and logical operators. (b) PrexSyn generates molecules from Enamine building blocks ($n = 480k$) and reaction templates ($n = 115$). (c) Leads are diversified via modification algorithms, verified by AiZynthFinder, and ranked by hit rate, diversity, and expansion.

Experimental Setup & Baselines

Baseline. Repeated PrexSyn sampling: N additional draws from the same property specification, subject to identical evaluation. This tests whether adding a modification layer outperforms simply sampling more from the generative model.

Hit classification. A variant is a hit if it is (a) synthesizable (AiZynthFinder route found) and (b) property-conserving on both axes below. Thresholds are parameters; results are reported at multiple stringency levels (e.g., Tanimoto ≥ 0.6, 0.7, 0.85; desirability ≥ 0.8).

Property conservation is assessed on two complementary axes. Substructural: ECFP4 Tanimoto similarity between variant and specification fingerprint. Physicochemical: per-descriptor deltas (MW ±50 Da, CLogP ±1.0, TPSA ±20 Å^2, HBD ±1, HBA ±2, rotatable bonds ±2) mapped to 0–1 desirability scores and aggregated by

geometric mean. 3D shape and electrostatic similarity via ESPsim [12] is reported as a supplementary diagnostic.

Primary metrics. (1) Hit rate: fraction of synthesizable variants meeting conservation thresholds. (2) Expansion factor: unique hits from modification / unique hits from baseline, the core measure of marginal value.

Secondary metrics. Chemical diversity among hits (1 − mean pairwise Tanimoto); wall-clock compute cost per specification; proximity to known bioactive space (Tanimoto to ChEMBL reference molecule, diagnostic only).

Complementarity. Pairwise Jaccard overlap between method hit sets; identification of optimal 2- and 3-method combinations by total unique-hit coverage.

Stratification. All metrics reported per baseline-quality bin to reveal when post-generation modification adds the most marginal value.

Timeline to Milestone 2

Week 1 (Mar 9–16): PrexSyn environment and inference setup (Tyler); AiZynthFinder integration (Jingxin); pilot saturation study, 5 specs × 4 methods × 1K variants, to determine diversity plateau (Yuma). *Bottleneck: GPU access for PrexSyn inference.*

Week 2 (Mar 16–23): Full benchmark: 100 stratified specs × 4 methods + repeated-sampling baseline. Ben leads baseline generation; all team members run parallel variant generation. *Bottleneck: AiZynthFinder throughput.*

Week 3 (Mar 23–30): Evaluation pipeline and complementarity analysis (Yuma, Tyler); figures and Milestone 2 draft (Ben, Jingxin).

Bottleneck mitigation: Spec count reduced to 100 via pre-stratified sampling; variant count capped by pilot saturation; canonical-SMILES deduplication before AiZynthFinder; parallel execution on rented multi-core compute (∼80K molecules, under 1 hour on 16 cores).

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